



January 30, 2025

Cindy Tomaszewski
City of Cadillac
1121 Plett Rd
Cadillac, MI 49601

RE: Project: Effluent PFAS
Pace Project No.: 50391678

Dear Cindy Tomaszewski:

Enclosed are the analytical results for sample(s) received by the laboratory on January 15, 2025. The results relate only to the samples included in this report. Results reported herein conform to the applicable TNI/NELAC Standards and the laboratory's Quality Manual, where applicable, unless otherwise noted in the body of the report.

The test results provided in this final report were generated by each of the following laboratories within the Pace Network:

- Pace Analytical Services - Minneapolis

If you have any questions concerning this report, please feel free to contact me.

Sincerely,

A handwritten signature in blue ink that reads 'Brian Hall'.

Brian Hall
brian.hall@pacelabs.com
(616)975-4500
Project Manager

Enclosures



REPORT OF LABORATORY ANALYSIS

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CERTIFICATIONS

Project: Effluent PFAS

Pace Project No.: 50391678

Pace Analytical Services, LLC - Minneapolis MN

1700 Elm Street SE, Minneapolis, MN 55414

Alabama Certification #: 40770

Alaska Contaminated Sites Certification #: 17-009

Alaska DW Certification #: MN00064

Arizona Certification #: AZ0014

Arkansas DW Certification #: MN00064

Arkansas WW Certification #: 88-0680

California Certification #: 2929

Colorado Certification #: MN00064

Connecticut Certification #: PH-0256

DoD Certification via A2LA #: 2926.01

EPA Region 8 Tribal Water Systems+Wyoming DW
Certification #: via MN 027-053-137

Florida Certification #: E87605

Georgia Certification #: 959

GMP+ Certification #: GMP050884

Hawaii Certification #: MN00064

Idaho Certification #: MN00064

Illinois Certification #: 200011

Indiana Certification #: C-MN-01

Iowa Certification #: 368

ISO/IEC 17025 Certification via A2LA #: 2926.01

Kansas Certification #: E-10167

Kentucky DW Certification #: 90062

Kentucky WW Certification #: 90062

Louisiana DEQ Certification #: AI-03086

Louisiana DW Certification #: MN00064

Maine Certification #: MN00064

Maryland Certification #: 322

Michigan Certification #: 9909

Minnesota Certification #: 027-053-137

Minnesota Dept of Ag Approval: via MN 027-053-137

Minnesota Petrofund Registration #: 1240

Mississippi Certification #: MN00064

Missouri Certification #: 10100

Montana Certification #: CERT0092

Nebraska Certification #: NE-OS-18-06

Nevada Certification #: MN00064

New Hampshire Certification #: 2081

New Jersey Certification #: MN002

New York Certification #: 11647

North Carolina DW Certification #: 27700

North Carolina WW Certification #: 530

North Dakota Certification (A2LA) #: R-036

North Dakota Certification (MN) #: R-036

Ohio DW Certification #: 41244

Ohio VAP Certification (1700) #: CL101

Oklahoma Certification #: 9507

Oregon Primary Certification #: MN300001

Oregon Secondary Certification #: MN200001

Pennsylvania Certification #: 68-00563

Puerto Rico Certification #: MN00064

South Carolina Certification #: 74003001

Tennessee Certification #: TN02818

Texas Certification #: T104704192

Utah Certification #: MN00064

Vermont Certification #: VT-027053137

Virginia Certification #: 460163

Washington Certification #: C486

West Virginia DEP Certification #: 382

West Virginia DW Certification #: 9952 C

Wisconsin Certification #: 999407970

Wyoming UST Certification via A2LA #: 2926.01

USDA Permit #: P330-19-00208

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SAMPLE SUMMARY

Project: Effluent PFAS

Pace Project No.: 50391678

Lab ID	Sample ID	Matrix	Date Collected	Date Received
50391678001	EFFLUENT	Water	01/14/25 12:55	01/15/25 11:22

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SAMPLE ANALYTE COUNT

Project: Effluent PFAS

Pace Project No.: 50391678

Lab ID	Sample ID	Method	Analysts	Analytes Reported	Laboratory
50391678001	EFFLUENT	EPA 1633 DRAFT	MJL	64	PASI-M

PASI-M = Pace Analytical Services - Minneapolis

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SUMMARY OF DETECTION

Project: Effluent PFAS

Pace Project No.: 50391678

Lab Sample ID	Client Sample ID					
Method	Parameters	Result	Units	Report Limit	Analyzed	Qualifiers
50391678001	EFFLUENT					
EPA 1633 DRAFT	PFBS	9.7	ng/L	4.7	01/29/25 03:08	
EPA 1633 DRAFT	PFHxA	6.8	ng/L	4.7	01/29/25 03:08	
EPA 1633 DRAFT	PFBA	3.1J	ng/L	18.8	01/29/25 03:08	
EPA 1633 DRAFT	PFOS	1.9J	ng/L	4.7	01/29/25 03:08	

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ANALYTICAL RESULTS

Project: Effluent PFAS

Pace Project No.: 50391678

Sample: EFFLUENT Lab ID: 50391678001 Collected: 01/14/25 12:55 Received: 01/15/25 11:22 Matrix: Water

Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
EPA 1633 DRAFT Water									
Analytical Method: EPA 1633 DRAFT Preparation Method: EPA 1633 DRAFT									
Pace Analytical Services - Minneapolis									
11CI-PF3OUdS	<3.7	ng/L	18.8	3.7	1	01/27/25 13:54	01/29/25 03:08	763051-92-9	
3:3 FTCA	<10.9	ng/L	23.5	10.9	1	01/27/25 13:54	01/29/25 03:08	356-02-5	
4:2 FTS	<2.3	ng/L	18.8	2.3	1	01/27/25 13:54	01/29/25 03:08	757124-72-4	
5:3 FTCA	<22.7	ng/L	117	22.7	1	01/27/25 13:54	01/29/25 03:08	914637-49-3	
6:2 FTS	<4.6	ng/L	18.8	4.6	1	01/27/25 13:54	01/29/25 03:08	27619-97-2	
7:3 FTCA	<24.6	ng/L	117	24.6	1	01/27/25 13:54	01/29/25 03:08	812-70-4	
8:2 FTS	<4.9	ng/L	18.8	4.9	1	01/27/25 13:54	01/29/25 03:08	39108-34-4	
9CI-PF3ONS	<3.9	ng/L	18.8	3.9	1	01/27/25 13:54	01/29/25 03:08	756426-58-1	
ADONA	<2.8	ng/L	18.8	2.8	1	01/27/25 13:54	01/29/25 03:08	919005-14-4	
HFPO-DA	<3.7	ng/L	18.8	3.7	1	01/27/25 13:54	01/29/25 03:08	13252-13-6	
NEtFOSAA	<1.6	ng/L	4.7	1.6	1	01/27/25 13:54	01/29/25 03:08	2991-50-6	
NEtFOSA	<1.2	ng/L	4.7	1.2	1	01/27/25 13:54	01/29/25 03:08	4151-50-2	
NEtFOSE	<15.1	ng/L	47.0	15.1	1	01/27/25 13:54	01/29/25 03:08	1691-99-2	
NFDHA	<2.7	ng/L	9.4	2.7	1	01/27/25 13:54	01/29/25 03:08	151772-58-6	
NMeFOSAA	<1.5	ng/L	4.7	1.5	1	01/27/25 13:54	01/29/25 03:08	2355-31-9	
NMeFOSA	<1.2	ng/L	4.7	1.2	1	01/27/25 13:54	01/29/25 03:08	31506-32-8	
NMeFOSE	<12.5	ng/L	47.0	12.5	1	01/27/25 13:54	01/29/25 03:08	24448-09-7	
PFBS	9.7	ng/L	4.7	1.5	1	01/27/25 13:54	01/29/25 03:08	375-73-5	
PFDA	<0.84	ng/L	4.7	0.84	1	01/27/25 13:54	01/29/25 03:08	335-76-2	
PFHxA	6.8	ng/L	4.7	0.71	1	01/27/25 13:54	01/29/25 03:08	307-24-4	
PFBA	3.1J	ng/L	18.8	2.9	1	01/27/25 13:54	01/29/25 03:08	375-22-4	
PFDS	<1.2	ng/L	4.7	1.2	1	01/27/25 13:54	01/29/25 03:08	335-77-3	
PFDoS	<1.3	ng/L	4.7	1.3	1	01/27/25 13:54	01/29/25 03:08	79780-39-5	
PFEESA	<1.5	ng/L	9.4	1.5	1	01/27/25 13:54	01/29/25 03:08	113507-82-7	
PFHpS	<1.1	ng/L	4.7	1.1	1	01/27/25 13:54	01/29/25 03:08	375-92-8	
PFMBA	<1.5	ng/L	9.4	1.5	1	01/27/25 13:54	01/29/25 03:08	863090-89-5	
PFMPA	<2.3	ng/L	9.4	2.3	1	01/27/25 13:54	01/29/25 03:08	377-73-1	
PFNS	<1.1	ng/L	4.7	1.1	1	01/27/25 13:54	01/29/25 03:08	68259-12-1	
PFOSA	<1.1	ng/L	4.7	1.1	1	01/27/25 13:54	01/29/25 03:08	754-91-6	
PFPeA	<1.4	ng/L	9.4	1.4	1	01/27/25 13:54	01/29/25 03:08	2706-90-3	
PFPeS	<0.86	ng/L	4.7	0.86	1	01/27/25 13:54	01/29/25 03:08	2706-91-4	
PFDoA	<1.0	ng/L	4.7	1.0	1	01/27/25 13:54	01/29/25 03:08	307-55-1	
PFHpA	<1.0	ng/L	4.7	1.0	1	01/27/25 13:54	01/29/25 03:08	375-85-9	
PFHxS	<1.2	ng/L	4.7	1.2	1	01/27/25 13:54	01/29/25 03:08	355-46-4	
PFNA	<1.1	ng/L	4.7	1.1	1	01/27/25 13:54	01/29/25 03:08	375-95-1	
PFOS	1.9J	ng/L	4.7	0.82	1	01/27/25 13:54	01/29/25 03:08	1763-23-1	
PFOA	<1.6	ng/L	4.7	1.6	1	01/27/25 13:54	01/29/25 03:08	335-67-1	
PFTeDA	<1.3	ng/L	4.7	1.3	1	01/27/25 13:54	01/29/25 03:08	376-06-7	
PFTrDA	<0.93	ng/L	4.7	0.93	1	01/27/25 13:54	01/29/25 03:08	72629-94-8	
PFUnA	<1.2	ng/L	4.7	1.2	1	01/27/25 13:54	01/29/25 03:08	2058-94-8	
Surrogates									
13C2-PFDoA (S)	77	%.	10-130		1	01/27/25 13:54	01/29/25 03:08		
13C3HFPO-DA (S)	91	%.	40-130		1	01/27/25 13:54	01/29/25 03:08		
13C3-PFBS (S)	96	%.	40-135		1	01/27/25 13:54	01/29/25 03:08		
13C3-PFHxS (S)	94	%.	40-130		1	01/27/25 13:54	01/29/25 03:08		

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ANALYTICAL RESULTS

Project: Effluent PFAS
Pace Project No.: 50391678

Sample: EFFLUENT Lab ID: 50391678001 Collected: 01/14/25 12:55 Received: 01/15/25 11:22 Matrix: Water

Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
EPA 1633 DRAFT Water									
Analytical Method: EPA 1633 DRAFT Preparation Method: EPA 1633 DRAFT									
Pace Analytical Services - Minneapolis									
Surrogates									
13C4-PFBA (S)	88	%.	5-130		1	01/27/25 13:54	01/29/25 03:08		
13C4-PFHpA (S)	87	%.	40-130		1	01/27/25 13:54	01/29/25 03:08		
13C5-PFHxA (S)	85	%.	40-130		1	01/27/25 13:54	01/29/25 03:08		
13C5-PFPeA (S)	88	%.	40-130		1	01/27/25 13:54	01/29/25 03:08		
13C6-PFDA (S)	86	%.	40-130		1	01/27/25 13:54	01/29/25 03:08		
13C8-PFOA (S)	84	%.	40-130		1	01/27/25 13:54	01/29/25 03:08		
13C8-PFOS (S)	89	%.	40-130		1	01/27/25 13:54	01/29/25 03:08		
13C8-PFOSA (S)	114	%.	40-130		1	01/27/25 13:54	01/29/25 03:08		
13C9-PFNA (S)	86	%.	40-130		1	01/27/25 13:54	01/29/25 03:08		
d3-MeFOSAA (S)	87	%.	40-170		1	01/27/25 13:54	01/29/25 03:08		
d3-NMeFOSA (S)	89	%.	10-130		1	01/27/25 13:54	01/29/25 03:08		
d5-EtFOSAA (S)	86	%.	25-135		1	01/27/25 13:54	01/29/25 03:08		
d5-NEtFOSA (S)	78	%.	10-130		1	01/27/25 13:54	01/29/25 03:08		
d7-NMeFOSE (S)	130	%.	10-130		1	01/27/25 13:54	01/29/25 03:08		
d9-NEtFOSE (S)	111	%.	10-130		1	01/27/25 13:54	01/29/25 03:08		
13C2-PFTA (S)	62	%.	10-130		1	01/27/25 13:54	01/29/25 03:08		
13C7-PFUdA (S)	82	%.	30-130		1	01/27/25 13:54	01/29/25 03:08		
13C24:2FTS (S)	116	%.	40-200		1	01/27/25 13:54	01/29/25 03:08		
13C26:2FTS (S)	115	%.	40-200		1	01/27/25 13:54	01/29/25 03:08		
13C28:2FTS (S)	110	%.	40-300		1	01/27/25 13:54	01/29/25 03:08		

REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Effluent PFAS

Pace Project No.: 50391678

QC Batch: 989426

Analysis Method: EPA 1633 DRAFT

QC Batch Method: EPA 1633 DRAFT

Analysis Description: 1633 W

Laboratory: Pace Analytical Services - Minneapolis

Associated Lab Samples: 50391678001

METHOD BLANK: 5166091

Matrix: Water

Associated Lab Samples: 50391678001

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
11CI-PF3OUdS	ng/L	<0.73	3.7	0.73	01/29/25 02:06	
3:3 FTCA	ng/L	<2.2	4.7	2.2	01/29/25 02:06	
4:2 FTS	ng/L	<0.45	3.7	0.45	01/29/25 02:06	
5:3 FTCA	ng/L	<4.5	23.4	4.5	01/29/25 02:06	
6:2 FTS	ng/L	<0.91	3.7	0.91	01/29/25 02:06	
7:3 FTCA	ng/L	<4.9	23.4	4.9	01/29/25 02:06	
8:2 FTS	ng/L	<0.98	3.7	0.98	01/29/25 02:06	
9CI-PF3ONS	ng/L	<0.77	3.7	0.77	01/29/25 02:06	
ADONA	ng/L	<0.55	3.7	0.55	01/29/25 02:06	
HFPO-DA	ng/L	<0.73	3.7	0.73	01/29/25 02:06	
NETFOSA	ng/L	<0.24	0.94	0.24	01/29/25 02:06	
NETFOSAA	ng/L	<0.33	0.94	0.33	01/29/25 02:06	
NETFOSE	ng/L	<3.0	9.4	3.0	01/29/25 02:06	
NFDHA	ng/L	<0.53	1.9	0.53	01/29/25 02:06	
NMeFOSA	ng/L	<0.25	0.94	0.25	01/29/25 02:06	
NMeFOSAA	ng/L	<0.30	0.94	0.30	01/29/25 02:06	
NMeFOSE	ng/L	<2.5	9.4	2.5	01/29/25 02:06	
PFBA	ng/L	<0.57	3.7	0.57	01/29/25 02:06	
PFBS	ng/L	<0.30	0.94	0.30	01/29/25 02:06	
PFDA	ng/L	<0.17	0.94	0.17	01/29/25 02:06	
PFDaA	ng/L	<0.21	0.94	0.21	01/29/25 02:06	
PFDoS	ng/L	<0.26	0.94	0.26	01/29/25 02:06	
PFDS	ng/L	<0.25	0.94	0.25	01/29/25 02:06	
PFEESA	ng/L	<0.30	1.9	0.30	01/29/25 02:06	
PFHpA	ng/L	<0.20	0.94	0.20	01/29/25 02:06	
PFHpS	ng/L	<0.22	0.94	0.22	01/29/25 02:06	
PFHxA	ng/L	<0.14	0.94	0.14	01/29/25 02:06	
PFHxS	ng/L	<0.24	0.94	0.24	01/29/25 02:06	
PFMBA	ng/L	<0.30	1.9	0.30	01/29/25 02:06	
PFMPA	ng/L	<0.45	1.9	0.45	01/29/25 02:06	
PFNA	ng/L	<0.22	0.94	0.22	01/29/25 02:06	
PFNS	ng/L	<0.23	0.94	0.23	01/29/25 02:06	
PFOA	ng/L	<0.32	0.94	0.32	01/29/25 02:06	
PFOS	ng/L	<0.16	0.94	0.16	01/29/25 02:06	
PFOSA	ng/L	<0.22	0.94	0.22	01/29/25 02:06	
PFPeA	ng/L	<0.28	1.9	0.28	01/29/25 02:06	
PFPeS	ng/L	<0.17	0.94	0.17	01/29/25 02:06	
PFTeDA	ng/L	<0.27	0.94	0.27	01/29/25 02:06	
PFTTrDA	ng/L	<0.19	0.94	0.19	01/29/25 02:06	
PFUnA	ng/L	<0.24	0.94	0.24	01/29/25 02:06	

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QUALITY CONTROL DATA

Project: Effluent PFAS

Pace Project No.: 50391678

METHOD BLANK: 5166091

Matrix: Water

Associated Lab Samples: 50391678001

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
13C2-PFDaA (S)	%.	82	10-130		01/29/25 02:06	
13C2-PFTA (S)	%.	83	10-130		01/29/25 02:06	
13C24:2FTS (S)	%.	105	40-200		01/29/25 02:06	
13C26:2FTS (S)	%.	106	40-200		01/29/25 02:06	
13C28:2FTS (S)	%.	107	40-300		01/29/25 02:06	
13C3-PFBS (S)	%.	93	40-135		01/29/25 02:06	
13C3-PFHxS (S)	%.	90	40-130		01/29/25 02:06	
13C3HFPO-DA (S)	%.	90	40-130		01/29/25 02:06	
13C4-PFBA (S)	%.	87	5-130		01/29/25 02:06	
13C4-PFHpA (S)	%.	89	40-130		01/29/25 02:06	
13C5-PFHxA (S)	%.	87	40-130		01/29/25 02:06	
13C5-PFPeA (S)	%.	87	40-130		01/29/25 02:06	
13C6-PFDA (S)	%.	89	40-130		01/29/25 02:06	
13C7-PFUDa (S)	%.	85	30-130		01/29/25 02:06	
13C8-PFOA (S)	%.	82	40-130		01/29/25 02:06	
13C8-PFOS (S)	%.	88	40-130		01/29/25 02:06	
13C8-PFOSA (S)	%.	71	40-130		01/29/25 02:06	
13C9-PFNA (S)	%.	88	40-130		01/29/25 02:06	
d3-MeFOSAA (S)	%.	86	40-170		01/29/25 02:06	
d3-NMeFOSA (S)	%.	71	10-130		01/29/25 02:06	
d5-EtFOSAA (S)	%.	87	25-135		01/29/25 02:06	
d5-NEtFOSA (S)	%.	74	10-130		01/29/25 02:06	
d7-NMeFOSE (S)	%.	81	10-130		01/29/25 02:06	
d9-NEtFOSE (S)	%.	85	10-130		01/29/25 02:06	

LABORATORY CONTROL SAMPLE & LCSD: 5166092

5166093

Parameter	Units	Spike Conc.	LCS Result	LCSD Result	LCS % Rec	LCSD % Rec	% Rec Limits	RPD	Max RPD	Qualifiers
11CI-PF3OUdS	ng/L	88.2	75.2	80.0	85	91	55-160	6	30	
3:3 FTCA	ng/L	117	103	98.6	88	84	65-130	4	30	
4:2 FTS	ng/L	87.6	85.7	85.1	98	97	70-145	1	30	
5:3 FTCA	ng/L	584	511	507	88	87	70-135	1	30	
6:2 FTS	ng/L	88.8	90.8	86.7	102	98	65-155	5	30	
7:3 FTCA	ng/L	584	500	503	86	86	50-145	1	30	
8:2 FTS	ng/L	89.9	93.4	90.9	104	101	60-150	3	30	
9CI-PF3ONS	ng/L	87.6	76.4	81.0	87	92	70-155	6	30	
ADONA	ng/L	88.2	78.1	82.3	89	93	65-145	5	30	
HFPO-DA	ng/L	93.4	82.2	87.6	88	94	70-140	6	30	
NEtFOSA	ng/L	23.4	21.9	21.6	94	92	65-145	1	30	
NEtFOSAA	ng/L	23.4	19.5	20.3	83	87	70-145	4	30	
NEtFOSE	ng/L	234	229	224	98	96	70-135	2	30	
NFDHA	ng/L	46.7	44.3	49.3	95	105	50-150	11	30	
NMeFOSA	ng/L	23.4	22.0	21.7	94	93	60-150	1	30	
NMeFOSAA	ng/L	23.4	21.5	21.8	92	93	50-140	1	30	

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QUALITY CONTROL DATA

Project: Effluent PFAS

Pace Project No.: 50391678

LABORATORY CONTROL SAMPLE & LCSD:		5166092	5166093							
Parameter	Units	Spike Conc.	LCS Result	LCSD Result	LCS % Rec	LCSD % Rec	% Rec Limits	RPD	Max RPD	Qualifiers
NMeFOSE	ng/L	234	225	225	96	96	70-145	0	30	
PFBA	ng/L	93.4	87.7	89.5	94	96	70-140	2	30	
PFBS	ng/L	20.7	18.5	18.4	89	89	60-145	1	30	
PFDA	ng/L	23.4	22.6	22.3	97	95	70-140	1	30	
PFDoA	ng/L	23.4	27.0	22.3	116	95	70-140	19	30	
PFDoS	ng/L	22.7	18.9	19.5	83	86	50-145	3	30	
PFDS	ng/L	22.5	18.2	20.8	81	92	60-145	13	30	
PFEESA	ng/L	41.6	38.4	40.5	92	97	70-140	5	30	
PFHpA	ng/L	23.4	21.6	22.4	92	96	70-150	4	30	
PFHpS	ng/L	22.2	20.0	20.5	90	92	70-150	3	30	
PFHxA	ng/L	23.4	20.2	22.2	87	95	70-145	9	30	
PFHxS	ng/L	21.4	18.8	17.8	88	83	65-145	5	30	
PFMBA	ng/L	46.7	41.9	42.4	90	91	60-150	1	30	
PFMPA	ng/L	46.7	42.8	43.1	92	92	55-140	1	30	
PFNA	ng/L	23.4	21.7	21.3	93	91	70-150	2	30	
PFNS	ng/L	22.5	20.2	21.0	90	93	65-145	4	30	
PFOA	ng/L	23.4	21.7	22.1	93	94	70-150	2	30	
PFOS	ng/L	21.7	18.7	19.0	86	88	55-150	2	30	
PFOSA	ng/L	23.4	21.7	21.6	93	92	70-145	0	30	
PFPeA	ng/L	46.7	43.8	43.4	94	93	65-135	1	30	
PFPeS	ng/L	22	20.6	20.8	94	94	65-140	1	30	
PFTeDA	ng/L	23.4	21.8	22.3	93	95	60-140	2	30	
PFTrDA	ng/L	23.4	27.4	22.5	117	96	65-140	19	30	
PFUnA	ng/L	23.4	21.9	21.7	94	93	70-145	1	30	
13C2-PFDoA (S)	%				70	82	10-130			
13C2-PFTA (S)	%				84	79	10-130			
13C24:2FTS (S)	%				102	98	40-200			
13C26:2FTS (S)	%				105	105	40-200			
13C28:2FTS (S)	%				106	104	40-300			
13C3-PFBS (S)	%				98	94	40-135			
13C3-PFHxS (S)	%				98	94	40-130			
13C3HFPO-DA (S)	%				94	90	40-130			
13C4-PFBA (S)	%				90	88	5-130			
13C4-PFHpA (S)	%				93	87	40-130			
13C5-PFHxA (S)	%				94	86	40-130			
13C5-PFPeA (S)	%				92	89	40-130			
13C6-PFDA (S)	%				91	86	40-130			
13C7-PFUdA (S)	%				90	87	30-130			
13C8-PFOA (S)	%				90	87	40-130			
13C8-PFOS (S)	%				97	93	40-130			
13C8-PFOSA (S)	%				83	82	40-130			
13C9-PFNA (S)	%				92	91	40-130			
d3-MeFOSAA (S)	%				93	90	40-170			
d3-NMeFOSA (S)	%				77	73	10-130			
d5-EtFOSAA (S)	%				90	89	25-135			
d5-NEtFOSA (S)	%				78	74	10-130			
d7-NMeFOSE (S)	%				84	82	10-130			

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Effluent PFAS

Pace Project No.: 50391678

LABORATORY CONTROL SAMPLE & LCSD: 5166092

5166093

Parameter	Units	Spike Conc.	LCS Result	LCSD Result	LCS % Rec	LCSD % Rec	% Rec Limits	RPD	Max RPD	Qualifiers
d9-NEtFOSE (S)	%.				87	86	10-130			

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 5166095

5166096

Parameter	Units	10721344005 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
11CI-PF3OUdS	ng/L	ND	439	419	391	355	89	85	50-150	10	30	
3:3 FTCA	ng/L	ND	581	556	200	189	34	34	50-150	6	30	M1
4:2 FTS	ng/L	ND	436	417	419	381	96	92	50-150	9	30	
5:3 FTCA	ng/L	ND	2900	2780	2650	2660	91	96	50-150	0	30	
6:2 FTS	ng/L	ND	441	422	426	393	96	93	50-150	8	30	
7:3 FTCA	ng/L	ND	2900	2780	2560	2610	88	94	50-150	2	30	
8:2 FTS	ng/L	ND	447	428	451	416	101	97	50-150	8	30	
9CI-PF3ONS	ng/L	ND	436	417	409	363	94	87	50-150	12	30	
ADONA	ng/L	ND	439	419	391	343	89	82	50-150	13	30	
HFPO-DA	ng/L	ND	465	444	413	374	89	84	50-150	10	30	
NEtFOSA	ng/L	ND	116	111	101	94.9	87	85	50-150	6	30	
NEtFOSAA	ng/L	ND	116	111	89.5	90.9	77	82	50-150	2	30	
NEtFOSE	ng/L	ND	1160	1110	1070	1020	93	92	50-150	5	30	
NFDHA	ng/L	ND	232	222	211	208	91	93	50-150	2	30	
NMeFOSA	ng/L	ND	116	111	108	98.2	93	88	50-150	9	30	
NMeFOSAA	ng/L	ND	116	111	108	93.7	93	84	50-150	14	30	
NMeFOSE	ng/L	ND	1160	1110	1050	993	91	89	50-150	6	30	
PFBA	ng/L	20.4	465	444	426	387	87	82	50-150	10	30	
PFBS	ng/L	ND	103	98.6	87.4	89.9	83	89	50-150	3	30	
PFDA	ng/L	ND	116	111	103	98.6	88	88	50-150	4	30	
PFDoA	ng/L	ND	116	111	108	101	93	91	50-150	7	30	
PFDoS	ng/L	ND	113	108	81.0	84.6	72	78	50-150	4	30	
PFDS	ng/L	ND	112	107	91.8	88.7	82	83	50-150	3	30	
PFEESA	ng/L	ND	207	198	184	174	89	88	50-150	6	30	
PFHpA	ng/L	5.3	116	111	109	101	90	86	50-150	8	30	
PFHpS	ng/L	ND	111	106	96.2	88.8	87	84	50-150	8	30	
PFHxA	ng/L	6.9	116	111	110	101	89	85	50-150	8	30	
PFHxS	ng/L	ND	106	102	92.7	83.3	84	78	50-150	11	30	
PFMBA	ng/L	ND	232	222	215	204	92	92	50-150	5	30	
PFMPA	ng/L	ND	232	222	56.8	55.1	24	25	50-150	3	30	M1
PFNA	ng/L	ND	116	111	109	96.8	93	86	50-150	12	30	
PFNS	ng/L	ND	112	107	95.0	90.9	85	85	50-150	4	30	
PFOA	ng/L	5.7	116	111	107	101	88	85	50-150	7	30	
PFOS	ng/L	ND	108	103	90.3	85.3	80	78	50-150	6	30	
PFOSA	ng/L	ND	116	111	98.4	98.2	85	88	50-150	0	30	
PFPeA	ng/L	ND	232	222	219	208	90	89	50-150	5	30	
PFPeS	ng/L	ND	109	104	98.4	89.1	90	85	50-150	10	30	
PFTeDA	ng/L	ND	116	111	103	98.5	89	89	50-150	5	30	
PFTTrDA	ng/L	ND	116	111	106	98.3	91	88	50-150	8	30	

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QUALITY CONTROL DATA

Project: Effluent PFAS

Pace Project No.: 50391678

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 5166095 5166096											
Parameter	Units	10721344005 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	Max RPD	Qual
PFUnA	ng/L	ND	116	111	105	98.7	90	89	50-150	6	30
13C2-PFDoA (S)	%						88	86	10-130		
13C2-PFTA (S)	%						86	86	10-130		
13C24:2FTS (S)	%						98	100	40-200		
13C26:2FTS (S)	%						93	95	40-200		
13C28:2FTS (S)	%						83	85	40-300		
13C3-PFBS (S)	%						103	99	40-135		
13C3-PFHxS (S)	%						102	100	40-130		
13C3HFPO-DA (S)	%						100	97	40-130		
13C4-PFBA (S)	%						8	8	5-130		
13C4-PFHpA (S)	%						93	91	40-130		
13C5-PFHxA (S)	%						98	92	40-130		
13C5-PFPeA (S)	%						85	81	40-130		
13C6-PFDA (S)	%						97	91	40-130		
13C7-PFUDa (S)	%						94	91	30-130		
13C8-PFOA (S)	%						95	93	40-130		
13C8-PFOS (S)	%						104	97	40-130		
13C8-PFOSA (S)	%						92	84	40-130		
13C9-PFNA (S)	%						93	92	40-130		
d3-MeFOSAA (S)	%						98	92	40-170		
d3-NMeFOSA (S)	%						85	82	10-130		
d5-EtFOSAA (S)	%						99	89	25-135		
d5-NEtFOSA (S)	%						86	85	10-130		
d7-NMeFOSE (S)	%						86	84	10-130		
d9-NEtFOSE (S)	%						90	86	10-130		

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QUALIFIERS

Project: Effluent PFAS
Pace Project No.: 50391678

DEFINITIONS

DF - Dilution Factor, if reported, represents the factor applied to the reported data due to dilution of the sample aliquot.
ND - Not Detected at or above adjusted reporting limit.
TNTC - Too Numerous To Count
J - Estimated concentration above the adjusted method detection limit and below the adjusted reporting limit.
MDL - Adjusted Method Detection Limit.
PQL - Practical Quantitation Limit.
RL - Reporting Limit - The lowest concentration value that meets project requirements for quantitative data with known precision and bias for a specific analyte in a specific matrix.
S - Surrogate
1,2-Diphenylhydrazine decomposes to and cannot be separated from Azobenzene using Method 8270. The result for each analyte is a combined concentration.
Consistent with EPA guidelines, unrounded data are displayed and have been used to calculate % recovery and RPD values.
LCS(D) - Laboratory Control Sample (Duplicate)
MS(D) - Matrix Spike (Duplicate)
DUP - Sample Duplicate
RPD - Relative Percent Difference
NC - Not Calculable.
SG - Silica Gel - Clean-Up
U - Indicates the compound was analyzed for, but not detected.
N-Nitrosodiphenylamine decomposes and cannot be separated from Diphenylamine using Method 8270. The result reported for each analyte is a combined concentration.
Reported results are not rounded until the final step prior to reporting. Therefore, calculated parameters that are typically reported as "Total" may vary slightly from the sum of the reported component parameters.
Pace Analytical is TNI accredited. Contact your Pace PM for the current list of accredited analytes.
TNI - The NELAC Institute.

ANALYTE QUALIFIERS

M1 Matrix spike recovery exceeded QC limits. Batch accepted based on laboratory control sample (LCS) recovery.

REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA CROSS REFERENCE TABLE

Project: Effluent PFAS
Pace Project No.: 50391678

Lab ID	Sample ID	QC Batch Method	QC Batch	Analytical Method	Analytical Batch
50391678001	EFFLUENT	EPA 1633 DRAFT	989426	EPA 1633 DRAFT	990079

REPORT OF LABORATORY ANALYSIS

CHAIN-OF-CUSTODY Analytical Request Document

Chain-of-Custody is a LEGAL DOCUMENT - Complete all relevant fields

[illegible]



Sample Conditions Upon Receipt Form (SCUR)

Date/Time: <u>1/15 1221</u>		Evaluated By: <u>BJH</u>		W0# : 50391678	
Client: <u>City of Cadillac</u>		PM: <u>BJH</u>		PM: BJH Due Date: 02/13/25 CLIENT: GR-Cadillac	
Lab Notified of Rush or Short Holds: YES NO <u>NA</u>					
Project Received Via: FedEx <u>UPS</u> Client Pace Courier Other: _____					Comments:
Custody Seal Present and Intact:		YES	NO	<u>NA</u>	
Received Sample Information Form (SIF): Drinking Waters Only		YES	NO	<u>NA</u>	
Short Hold Present (≤ 48 Hours):		YES	NO		
Sample Received in Hold:		<u>YES</u>	NO		
Custody Signature Present:		<u>YES</u>	NO		
Collector Signature Present:		<u>YES</u>	NO		
Sample Collected Today and On Ice:		YES	NO	<u>N/A</u>	
IR Gun #: <u>350</u> <u>351</u> Therm #: <u>282</u> <u>283</u>		Temp. should be 0°C - 6°C (Initial/Corrected)			
Ice Type: <u>WET Bagged / WET Loose</u> <u>BLUE</u> <u>NONE</u>		1. Cooler Temp. Upon Receipt: <u>1.9/2.1</u> °C			
Ice Location: <u>TOP</u> <u>BOTTOM</u> <u>MIDDLE</u> <u>DISPERSED</u>		2. Cooler Temp. Upon Receipt: _____ °C			
Temp Blank Received:		<u>YES</u>	NO		
Sample Label Matches COC (ID/Date/Time):		<u>YES</u>	NO		
Container Intact:		<u>YES</u>	NO		
Correct Container:		<u>YES</u>	NO		
Sufficient Volume:		<u>YES</u>	NO		
Sample pH Acceptable: All containers needing preservation are found to be in compliance with EPA recommendation. Denote with red dot on container lid if unacceptable. pH Strip Lot #: _____ Exceptions are VOA, coliform, LLHg, O&G/TPH, or any container with a septum cap or preserved with HCl		YES	NO	<u>N/A</u>	
Residual Chlorine Absent: <u>Cl₂ Strip Lot #: _____</u> Applies to SVOC 625, PCB/Pest. 608, Total/Amenable/Available/Free Cyanide		YES	NO	<u>N/A</u>	
VOA Headspace Acceptable (<6mm): Denote with silver x on rim of container cap if unacceptable.		YES	NO	<u>N/A</u>	
Trip Blank Received: <u>HCl</u> <u>MeOH</u> Other: _____		YES	NO	<u>ON HOLD</u>	
Comments:		3. Cooler Temp. Upon Receipt: _____ °C			
		4. Cooler Temp. Upon Receipt: _____ °C			
		Non-Conformance Form Required: YES <u>NO</u>			